

Analysis Plan: Ventricle Shape vs. Size in Schizophrenia

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Question

Enlarged lateral ventricles are one of the most replicated structural findings in schizophrenia (ENIGMA: Cohen's $d = 0.37$; van Erp et al., 2016), but the finding rests on a single volume number, and patient and control distributions overlap heavily. In a small twin study, ventricle shape revealed group differences more sensitively than volume (Styner et al., 2005), and ENIGMA's large subcortical shape meta-analysis did not include the ventricles (Gutman et al., 2022). This project tests, in a larger independent sample, whether lateral ventricle *shape* distinguishes people with schizophrenia from healthy controls better than ventricle *size*, and where on the ventricle any differences lie.

Data

OpenNeuro ds000030, UCLA Consortium for Neuropsychiatric Phenomics (Poldrack et al., 2016), FreeSurfer 6.0 derivatives: 50 people with schizophrenia (age 36.5 ± 8.9 , 76% male) and 125 healthy controls (31.5 ± 8.8 , 53% male), scanned on two 3T scanners. Public, de-identified data.

Surface models

Lateral ventricle labels (4, 43) are combined with choroid plexus labels (31, 63), so that choroid plexus differences cannot appear as shape differences. Each side is cleaned identically (morphological closing, radius 1 voxel; largest connected component) and converted to a closed surface in 3D Slicer.

Size

Ventricle volume per side, plus an ENIGMA-comparable volume (labels 4 + 43 only), adjusted for estimated total intracranial volume.

Shape

Procrustes-style alignment (iterative closest point with scaling) removes position, orientation, and size. Each surface is compared to a control template built from the other controls (leave-one-out, to avoid circularity). Mean surface distance is summarized in three regions per side (frontal horn, body, posterior horn), giving six shape features.

Comparison

(1) Effect sizes (Cohen's d) for size and for each shape region. (2) Cross-validated logistic regression using size only, shape only, and both, with templates rebuilt inside each training fold. Because groups are unequal, performance is reported as balanced accuracy and AUC; significance is tested against a label-permutation null (1,000 permutations).

Confounds

Patients are older, more often male, and more often scanned on the second scanner. Age, sex, intracranial volume, and scanner are included as covariates, and results are checked within each scanner separately.

Limitations

Antipsychotic exposure cannot be fully controlled. At 1 mm resolution, parts of the ventricle are about one voxel thick, producing small topological holes (Figure 1). The sample is modest, so the number of features is kept small.

Visualization

Distance maps on 3D ventricle models rendered in Blender, and 3D-printed average control and average patient ventricles.

References

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- Styner M, Lieberman JA, McClure RK, et al. Morphometric analysis of lateral ventricles in schizophrenia and healthy controls regarding genetic and disease-specific factors. *PNAS*. 2005;102(13):4872-4877.
- van Erp TGM, Hibar DP, Rasmussen JM, et al. Subcortical brain volume abnormalities in 2028 individuals with schizophrenia and 2540 healthy controls via the ENIGMA consortium. *Molecular Psychiatry*. 2016;21(4):547-553.

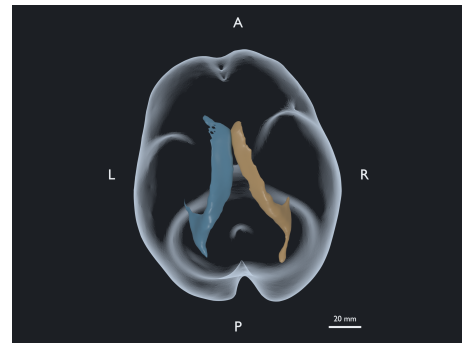


Figure 1. First reconstruction: lateral ventricles (including choroid plexus) of one healthy control (sub-10159), superior view, with brain outline for orientation. Holes at the anterior left ventricle mark regions about one voxel thick.